

ARITHMETIC AND ALGEBRA



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CHAPTER OVERVIEW

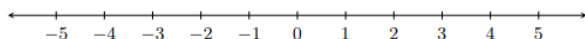
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1.1: Integers

We begin with a brief review of arithmetic with integers i.e. $\dots, -3, -2, -1, 0, 1, 2, 3, \dots$



Any number has a weight and a sign.

The **magnitude** (or **weight**) of a number is the distance it is from 0 on the number line.

✓ Example 1.1

For example, the weight of -5 is 5 and the weight of 7 is 7.

Two numbers are **opposites** if, on the number line, they are on opposite sides of zero, but the same distance away from zero.

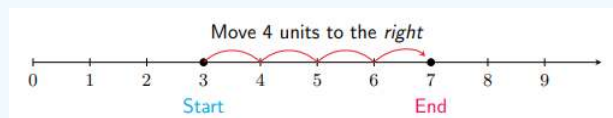
So, -5 is the opposite of 5, and 7 is the opposite of -7 and so on.

Addition

We can add two numbers with the help of a number line.

✓ Example 1.2

Adding two positive numbers: For example to add $3 + 4$, we start with 3 on the number line then move 4 units to the right. We land at 7 which is our answer.



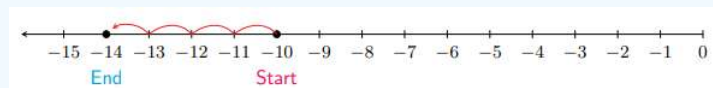
So, $3 + 4 = 7$. Notice that the answer has the same sign as the signs of 3 and 4 (both positive) and its weight comes from adding the weights of 3 and 4.

You always move to the right when you add a positive number

✓ Example 1.3

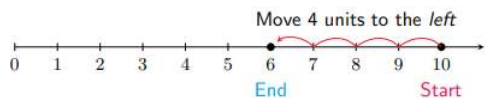
Adding two negative numbers: For example, $-10 + (-4)$ means you are adding a debt of \$4 to an already existing debt of \$10. So, we start at -10 on the number line and move 4 units to the left, to land at -14, which is the answer.

You always move to the left when you add a negative number (a debt)



So, $-10 + (-4) = -14$. Notice that the answer has the same sign as the signs of -10 and -4 (both negative) and its weight comes from adding the weights of -10 and -4

To add numbers of opposite signs, that is, a positive and a negative number, we can also use the number line. For example, to perform $10 + (-4)$, we start at 10 on the number line and then move 4 units to the left. We land at 6, which is the answer. Think of $10 + (-4)$ as having \$10 and adding a \$4 debt. Because we are adding a debt, we move to the left on the number line!

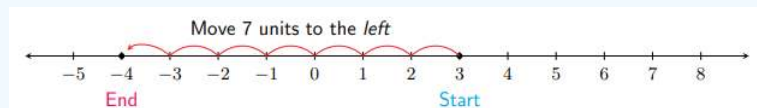


So, $10 + (-4) = 6$. Notice that the answer has the same sign as the sign of 10 (positive) because it is the number of larger weight, and its weight comes from finding the difference of the weights of 10 and -4

Notice that because we were adding two numbers of opposite signs, the answer ended up being the difference in weight (6) along with the sign of the number of larger weight (positive).

✓ Example 1.4

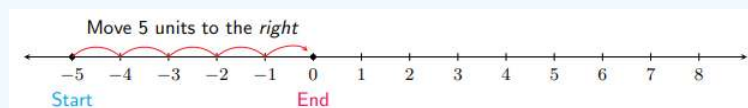
Adding two numbers of opposite signs: For example, $3 + (-7)$. We start at 3 and move toward 7 units to the left, and we land at -4, which is our answer.



So, $3 + (-7) = -4$. Notice that the answer has the same sign as the signs of -7 (negative) because it is the number of larger weight, and its weight comes from finding the difference of the weights of 3 and -7.

✓ Example 1.5

Adding opposites: We start at -5 on the number line and jump to the right 5 units to land finally at 0. So $-5 + 5 = 0$.



📌 Note 1.6

Two opposite numbers are called a zero-pair, because adding them always results in 0.

So, -5 and 5 are a zero-pair.

📌 Adding Integers

1. To **add two numbers of the same sign**, add their weights and place it after the sign.
2. To **add two numbers of opposite signs**, find the difference of their weights and place it after the sign of the number with the greater weight.

✓ Example 1.7

Add:

- a. $-8 + 19 = 11$
- b. $-8 + 4 = -4$
- c. $6 + (-9) = -3$
- d. $7 + (-2) = 5$
- e. $(-4) + (-7) = -11$
- f. $8 + 7 = 15$

Note 1.8

While we can add in any order: $4 + 2 = 2 + 4$, it is sometimes convenient to add up all of the negative numbers and add up all of the positive numbers, and then add the results. There are also times when it is best to notice certain simplifications if the numbers are added in a different order.

For example

$$-5 + 4 + 5 + (-8) = -5 + (-8) + 4 + 5 \text{ (by reordering)}$$

so,

$$-5 + 4 + 5 + (-8) = -5 + (-8) + 4 + 5 = -13 + 9 = -4$$

We could have also simplified this by noting that (-5) and 5 are zero-pair, so we are left with $4 + (-8)$ which is -4 .

Example 1.9

We can calculate

$$(-4) + (-5) + 7 + (-3) = (-4) + (-5) + (-3) + 7 = (-12) + 7 = -5$$

We could have simplified this by noting that (-4) and (-3) make -7 , and -7 and 7 are zero-pair, so the total is -5 .

Subtraction (as Addition of the Opposite)

Once we know how to add numbers, we are set to subtract numbers because subtraction is nothing but addition of the opposite. That is, subtracting $8 - 3$ (which reads: subtracting 3 from 8) is the same as $8 + (-3)$ (which reads: Adding -3 to 8.).

Example 1.10

So, $8 - 3 = 8 + (-3)$, and, we can use the rules of adding two numbers of opposite signs to find out that the answer is 5. We can also use the number line. We start at 8 and move 3 units to the left (adding -3 is adding a debt, so we move to the left). So $8 - 3 = 5$.
 5. - 4 - 3 - 2 - 1 0 1 2 3 4 5 6 7 8 End Start Move 3 units to the left

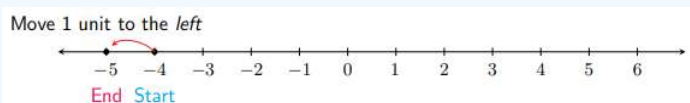
Example 1.11

To calculate $3 - 7$ we first rewrite it as an addition problem. $3 - 7 = 3 + (-7)$. We can either use the number line, or the rules of adding two numbers of opposite signs. And, $3 - 7 = 3 + (-7) = -4$

Example 1.12

To calculate $-4 - 1$, we first rewrite it as an addition problem. $-4 - 1 = -4 + (-1)$. We can either use the number line, or the rules of adding two numbers of the same signs. If we want to use the rules, both numbers are negative, so our answer will be negative, and, adding the weights of -4 and -1 is -5 . $-4 - 1 = -4 + (-1) = -5$

On the number line, we start at -4 and move 1 unit to the left, to land on -5 which is our answer.



Changing the subtractions to additions in this way is particularly useful when adding or subtracting several numbers (because we can add in any order).

✓ Example 1.13

$$\begin{aligned}
 & -3 - 7 + 5 + 7 + 13 - 6 - (-9) \\
 &= -3 + (-7) + 5 + 7 + 13 + (-6) + 9 \\
 &= -3 + (-7) + (-6) + 5 + 7 + 13 + 9 \\
 &= 18 \\
 & -3 - 7 + 5 + 7 + 13 - 6 - (-9) \\
 &= -3 + (-7) + 5 + 7 + 13 + (-6) + 9 \\
 &= 5 + 13 \\
 &= 18
 \end{aligned}$$

📌 Remark 1.14

Warning: The symbol "-" is used in two different ways. When it is between two expressions, it means subtract (e.g., $3 - 4$). Otherwise, it means 'opposite' or 'negative' (e.g., $-3 + 4$). So in the expression $-4 - (-3)$, the first and last "-" means opposite and the one in the middle one means subtract. The importance of understanding this can not be overestimated.

Multiplication and Division of Positive Numbers

Multiplication of integers is adding in the sense that $3 \times 4 = 4 + 4 + 4$

To multiply larger numbers, it is better to use the usual scheme of multiplication. For example:

✓ Example 1.15

Let us multiply 152 by 34. We will for convenience sake put the smaller number on the bottom (though it is not necessary). We have

$$\begin{array}{r}
 152 \\
 \times 34 \\
 \hline
 608 \\
 + 4560 \\
 \hline
 5168
 \end{array}$$

And division is the opposite of multiplication in the sense that to compute $45 \div 9$ is to find a number so that when we multiply by 9 we get 45. We run through our multiplication tables (which are hopefully in our head) to discover that 5 does the trick: $5 \times 9 = 45$ so that $45 \div 9 = 5$. We will discuss division from a different point of view when we discuss fractions.

To divide larger numbers, we can use long division. For example, let's divide 3571 by 11.

✓ Example 1.16

$$\begin{array}{r}
 324 \\
 11 \overline{) 3571} \\
 \underline{-33} \\
 271 \\
 \underline{-22} \\
 51 \\
 \underline{-44} \\
 7 = \text{remainder}
 \end{array}$$

Multiplication involving negative numbers

Multiplication is a little tricky to understand without the notion of distribution (discussed later). We will begin with noting again what it means to multiply a number by a positive number: So if we want to compute $4 \cdot (-7)$ we note

$$4 \cdot (-7) = (-7) + (-7) + (-7) + (-7) = -28$$

Note that since $4 \cdot 7 = 28$, $4 \cdot (-7) = -(4 \cdot 7)$. We can multiply positive numbers in any order: $4 \cdot 7 = 7 \cdot 4$. The same is true of positive and negative numbers:

$$(-7) \cdot 4 = 4 \cdot (-7) = -(4 \cdot 7) = -28$$

✓ Example 1.17

$$5 \cdot (-12) = -(5 \cdot 12) = -60 \text{ and } (-3) \cdot (-2) = -(3 \cdot (-2)) = -(- (3 \cdot 2)) = 6$$

✓ Example 1.18

So the size of the product of two numbers is the product of their sizes. The sign is positive if the signs are the same and negative if they are different.

✓ Example 1.19

Two quantities right next to each other, with no symbol between them (except for parentheses around either or both numbers), has an implicit multiplication. For example, $3(2) = 3 \times 2$.

✓ Example 1.20

Multiply:

- a. $(-5)(-8) = 40$
- b. $(-6) \cdot 7 = -42$
- c. $4 \cdot 12 = 48$
- d. $(-3)(-6) \cdot 4(-3) = 18 \cdot 4(-3) = 72(-3) = -216$ (multiplying from left to right)
- e. $(-3)(-5) \cdot 4(-2) = (-3) \cdot 4 \cdot (-5)(-2) = -12 \cdot 10 = -120$ (since we can multiply in any order it is convenient to see that $-5 \cdot -2 = 10$.)

Exponents of integers

Recall that a positive exponent represents the number of times a number is multiplied by itself.

✓ Example 1.21

Evaluate:

- a. $5^2 = 5 \cdot 5 = 25$
- b. $(-4)^3 = (-4) \cdot (-4) \cdot (-4) = 16 \cdot (-4) = -64$
- c. $(-7)^1 = -7$
- d. $-2^4 = -2 \cdot 2 \cdot 2 \cdot 2 = -16$ **Note:** The exponent here is for 2 not -2!
- e. $(-3)^4 = (-3) \cdot (-3) \cdot (-3) \cdot (-3) = 9 \cdot (-3) \cdot (-3) = -27 \cdot (-3) = 81$
- f. $(-2)^5 = (-2) \cdot (-2) \cdot (-2) \cdot (-2) \cdot (-2) = -32$

Exponent rules in details will be further discussed in Chapter 5.

Division involving negative numbers

Division is just a question of knowing multiplication and therefore has the same rule: The size of the quotient of two numbers is the quotient of the sizes. The sign is positive if the signs are the same and negative if they are different.

✓ Example 1.22

Divide:

- a. $(-42) \div 7 = -6$
- b. $81 \div (-9) = -9$

- c. $(-35) \div (-7) = 5$
- d. $14 \div 2 = 7$
- e. $0 \div 5 = 0$. Note When dividing 0 by any number, the answer is always 0.
- f. $-10 \div 0 = \text{undefined}$.

Note 1.23

Any number divided by 0 is undefined!

Multiplying and Dividing Integers

Consider two numbers at a time.

1. If the signs of the two numbers are the same, then the sign of the answer is positive.
2. If the signs of the two numbers are different, then the sign of the answer is negative.

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1.3: The Order of Operations

What is the meaning of the expression '3 times 4 plus 5'. Some will answer 17 while others may answer 27. Why? To take the ambiguity out, we can write

$$(3 \times 4) + 5 = 17$$

and

$$3 \cdot (4 + 5) = 27,$$

where we must first evaluate the quantity in parentheses. Since it can be somewhat cumbersome to write a lot of parentheses, there is an important convention or agreement that if we just write $3 \times 4 + 5$ we mean $(3 \times 4) + 5$. That is, in the absence of parentheses, we should multiply before we add. This is part of what is called The Order of Operations. This must be remembered.

Definition: 1.24: The Order of Operation

When evaluating an expression involving addition, subtraction, multiplication and division which has no parentheses or exponents, we first perform, from left to right, all of the multiplications and divisions. Then, from left to right, the additions and subtractions. If there are parts of the expression set off by parentheses, what is within the parentheses must be evaluated first.

Remark 1.25

Subtraction can be turned into addition and then addition can be done in any order, not necessarily from left to right. This explains why addition and subtraction come together in the order of operations. There will be a similar statement for multiplication and division but will be postponed until fractions are discussed.

'PE(MD)(AS)' is an easy way to remember the order of operations. This means that the order is: Parentheses, Exponents (this will be incorporated later), Multiplication and Division (taken together from left to right), and finally, Addition and Subtraction (taken together from left to right).

Let us try a few problems.

Example 1.26

- a. $3 + 2(3 + 5) = 3 + 2(8) = 3 + 16 = 19$
- b. $3 - 2(-4 + 7) = 3 - 2(3) = 3 - 6 = -3$
- c. $-3 - 4 - 2(-2 \cdot 6 - 5) = -3 - 4 - 2(-12 - 5) = -3 - 4 - 2(-17) = -3 - 4 - (-34) = -3 - 4 + 34 = 27$
- d. $-(3 - (-6)) - (1 - 4 \cdot (-5) + 4) = -(3 + 6) - (1 - (-20) + 4) = -9 - (1 + 20 + 4) = -9 - 25 = -9 + (-25) = -34$
- e. $-2(-14 \div 7 + 7) = -2(-2 + 7) = -2(5) = -10$
- f. $-3(-2 \cdot 7 - (-5)(4) \div 2) = -3(-14 - (-20) \div 2) = -3(-14 - (-10)) = -3(-4) = 12$
- g. $6 \div 2 \times 3 = 3 \times 3 = 9$ **Note:** $6 \div 2 \times 36 = 6 \div 6 = 1$
- h. $-2(3 - 1)2 - (8 - 22) \div 4 = -2(2)2 - (8 - 4) \div 4 = -2(4) - 4 \div 4 = -8 - 1 = -9$

Exit Problem

Evaluate: $(3^3 + 5) \div 4 - 4(7 - 2)$

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1.4: Fractions

In our previous section we identified integers: $0, \pm 1, \pm 2, \pm 3, \dots$. To this set, now we will add all ratios of integers with non-zero denominators, like $\frac{2}{7}, \frac{-11}{17}, \dots$. We call this the set of **rational numbers**. Any rational number looks like $\frac{p}{q}$ where p and q are integers and q is not 0.

Just as we are able to perform arithmetic operations with integers we can also perform arithmetic operations with rational numbers (fractions). The two types of fractions we will encounter are called proper and improper:

- **Proper fractions** have value less than 1, for example $\frac{2}{5}$ and $\frac{1}{8}$. Observe that for these fractions the numerator is less than the denominator.
- **Improper fractions** have value greater than or equal to 1, for example $\frac{7}{6}$ and $\frac{3}{2}$. For these fractions the numerator is greater than the denominator.

Each fractional value can have many different, equivalent forms, for example $1 = \frac{2}{2} = \frac{-5}{-5} = \dots$. In order to determine whether two fractions are equivalent we can use the fundamental principle of fractions.

The Fundamental Principle of Fractions

$$\frac{2}{3} = \frac{2 \cdot 4}{3 \cdot 4} = \frac{8}{12}$$

That is, as long as you multiply both numerator and denominator by the **same** number, the fraction value does not change, and you obtain equivalent fractions.

✓ Example 2.1

Write a fraction that is equivalent to $\frac{3}{5}$.

Solution

Begin with our original fraction $\frac{3}{5}$ and apply the fundamental principle of fractions to get

$$\frac{3}{5} = \frac{3 \cdot 2}{5 \cdot 2} = \frac{6}{10}$$

✓ Example 2.2

Simplify the fraction $\frac{15}{35}$.

Solution

Begin with our original fraction and apply the fundamental principle of fractions in reverse to get

$$\frac{15}{35} = \frac{3 \cdot 5}{7 \cdot 5} = \frac{3}{7}$$

Multiplying Fractions

We multiply numerators together and denominators together:

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{a \cdot c}{b \cdot d}$$

✓ Example 2.3

The product of these two fractions is done as follows:

$$\frac{14 \cdot 9}{3 \cdot 7} = \frac{2 \cdot 7 \cdot 3 \cdot 3}{3 \cdot 7} = \frac{6}{1} = 6$$

📌 The Reciprocal of a Fraction

The reciprocal of a fraction $\frac{p}{q}$ is the fraction formed by switching the numerator and denominator, namely $\frac{q}{p}$.

✓ Example 2.4

- The reciprocal of $\frac{3}{5}$ is $\frac{5}{3}$.
- The reciprocal of $\frac{-2}{7}$ is $\frac{7}{-2} = -\frac{7}{2}$.
- The reciprocal of $\frac{1}{8}$ is $\frac{8}{1} = 8$.
- The reciprocal of $4 = \frac{4}{1}$ is $\frac{1}{4}$.

Dividing Fractions

We multiply the first fraction by the **reciprocal** of the second:

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c} = \frac{a \cdot d}{b \cdot c}$$

✓ Example 2.5

The quotient of these two fractions is found as follows:

$$\frac{8}{3} \div \frac{4}{5} = \frac{8}{3} \cdot \frac{5}{4} = \frac{8 \cdot 5}{3 \cdot 4} = \frac{2 \cdot 4 \cdot 5}{3 \cdot 4} = \frac{10}{3}$$

Adding Fractions (with same denominators)

$$\frac{a}{b} + \frac{c}{b} = \frac{a + c}{b}$$

Subtracting Fractions (with same denominators)

$$\frac{a}{b} - \frac{c}{b} = \frac{a - c}{b}$$

✓ Example 2.6

Add $\frac{3}{5} + \frac{1}{5}$

Solution

$$\frac{3}{5} + \frac{1}{5} = \frac{3 + 1}{5} = \frac{4}{5}$$

Adding or Subtracting Fractions (with unlike denominators)

Adding (or subtracting) fractions with unlike denominators requires us to first find a common denominator. The LCD or least common denominator is the smallest number that both denominators evenly divide. Once we rewrite each of our fractions so their denominator is the LCD, we may add or subtract fractions according to the above properties.

📌 Finding the LCD

- Step 1: Make a list of (enough) multiples of each denominator.
- Step 2: Identify the lowest common multiple. If you can't see one, then your lists in Step 1. need to be expanded.

To be able to add or subtract fractions, we need to go one more step: Once you've identified the LCD, rewrite both fractions (by multiplying both numerator and denominator by the appropriate same number) to get the LCD as denominator.

✓ Example 2.7

Find the LCD and then add and simplify $\frac{3}{12} + \frac{5}{8}$.

Solution

Let us first find the LCD by following our procedure.

Step 1. Make a list of (enough) multiples:

$$8 : 8, 16, 24, 32, \dots$$

$$12 : 12, 24, 36, 48, \dots$$

Step 2. LCD: $24, 8 \cdot 3 = 24, 12 \cdot 2 = 24$

Step 3. Rewrite each fraction using the LCD:

$$\frac{3}{12} = \frac{3 \cdot 2}{12 \cdot 2} = \frac{6}{24}$$

and

$$\frac{5}{8} = \frac{5 \cdot 3}{8 \cdot 3} = \frac{15}{24}$$

Now we are ready to add our fractions

$$\frac{6}{24} + \frac{15}{24} = \frac{21}{24}$$

simplifying yields

$$1. \quad \frac{21}{24} = \frac{3 \cdot 7}{3 \cdot 8} = \frac{7}{8}$$

✓ Example 2.8

Find the LCD and then subtract and simplify $\frac{1}{9} - \frac{3}{5}$.

Solution

Let us first find the LCD by following our procedure.

Step 1. Make a list of (enough) multiples:

$$9 : 9, 18, 27, 36, 45, 54, 63, \dots$$

$$5 : 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, \dots$$

Step 2. LCD: 45

Step 3. Rewrite each fraction using the LCD:

$$\frac{1}{9} = \frac{1 \cdot 5}{9 \cdot 5} = \frac{5}{45}$$

and

$$\frac{3}{5} = \frac{3 \cdot 9}{5 \cdot 9} = \frac{27}{45}$$

Now we are ready to subtract our fractions, but, first, we rewrite the subtraction as addition of the opposite:

$$\frac{1}{9} - \frac{3}{5} = \frac{1}{9} + \left(-\frac{3}{5}\right) = \frac{5}{45} + \left(-\frac{27}{45}\right) = \frac{5 + (-27)}{45} = \frac{-22}{45}$$

Writing an Improper Fraction as a Mixed Number

1. Divide the numerator by the denominator.
2. If there is a remainder, write it over the denominator.

✓ Example 2.9

Write $\frac{42}{5}$ as a mixed number.

Solution

We begin by dividing the numerator 42 by the denominator 5 to get 8, with a remainder of 2. Our mixed number is $8\frac{2}{5}$.

Writing a Mixed Number as an Improper Fraction

1. Multiply the whole number and the denominator then add the numerator. Use the result as your new numerator.
2. The denominator remains the same

✓ Example 2.10

Write the mixed number $3\frac{5}{6}$ as an improper fraction.

Solution

1. Multiply the denominator by the whole number.
2. Add this result to the numerator.
3. Set this new numerator 23 over the denominator of 6.

$$3\frac{5}{6} = 3 + \frac{5}{6} = \frac{18}{6} + \frac{5}{6} = \frac{23}{6}$$

We multiply the whole number 3 and the denominator 6 to get 18. Next, we add to this the numerator 5 to get 23. This is our new numerator and our improper fraction becomes $\frac{23}{6}$.

Addition and Subtraction of Mixed Numbers

To add (or subtract) mixed numbers, we can convert the numbers into improper fractions, then add (or subtract) the fractions as we saw in this chapter.

✓ Example 1.4.1

Subtract $7 - 2\frac{3}{8}$.

Solution

First we convert $2\frac{3}{8} = \frac{19}{8}$. Then, we rewrite the subtraction operation as addition of opposite:

$$7 - \frac{19}{8} = 7 + \left(-\frac{19}{8}\right) = \frac{7}{1} + \left(-\frac{19}{8}\right) = \frac{56}{8} + \left(-\frac{19}{8}\right) = \frac{56 + (-19)}{8} = \frac{37}{8} = 4\frac{5}{8}$$

Also, we can keep mixed fractions and mixed fraction, and, add (or subtract) the integer parts together and the fraction parts together.

✓ Example 2.12

Add $7\frac{3}{4} + 3\frac{1}{5}$

Solution

Here, we add $7 + 3 = 10$ and $\frac{3}{4} + \frac{1}{5} = \frac{15}{20} + \frac{4}{20} = \frac{19}{20}$.

And, our final answer is $10\frac{19}{20}$. Note that $\frac{19}{20}$ is a proper fraction, so, our work is done. But, if our answer ended up with an improper fraction, we would have had to make the conversion to write the answer in simplified form.

Multiplying and Dividing of Mixed Numbers

Be careful when multiplying mixed numbers. You must first convert them to improper fractions and use the rules for multiplying fractions to finish your problem.

✓ Example 2.13

Multiply $2\frac{3}{5}$ and $3\frac{1}{2}$.

Solution

Begin by rewriting each mixed number as an improper fraction: $2\frac{3}{5} = \frac{13}{5}$ and $3\frac{1}{2} = \frac{7}{2}$. Now we proceed by multiplying the fractions

$$\frac{13}{5} \cdot \frac{7}{2} = \frac{13 \cdot 7}{5 \cdot 2} = \frac{91}{10}$$

We can now write the result (if we wish) as a mixed number: $9\frac{1}{10}$.

✓ Example 2.14

Divide $(1\frac{4}{5}) \div (1\frac{1}{2})$

Solution

Begin by rewriting each mixed number as a improper fraction: $1\frac{4}{5} = \frac{9}{5}$ and $1\frac{1}{2} = \frac{3}{2}$. Now we proceed by dividing the fractions

$$\frac{9}{5} \div \frac{3}{2} = \frac{9}{5} \cdot \frac{2}{3} = \frac{9 \cdot 2}{5 \cdot 3} = \frac{3 \cdot 2}{5 \cdot 1} = \frac{6}{5} = 1\frac{1}{5}$$

? Exit Problem

Evaluate: $\frac{3}{4} - 1\frac{5}{6}$.

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1.5: Decimal Numbers

Consider the number 23.7456. Each digit occupies a 'place'. The 2 is in the tens place, the three in the ones place, the 7 in the tenths place, the 4 in the hundredths place, the 5 in the thousandths place, and the 6 in the ten thousandths place. Why? Because:

$$23.7456 = 2 \cdot 10 + 3 \cdot 1 + 7 \cdot \frac{1}{10} + 4 \cdot \frac{1}{100} + 5 \cdot \frac{1}{1000} + 6 \cdot \frac{1}{10,000}$$

Rounding

Rounding is associated with cutting or truncating a number, and that the rounding compensates for the lost tail of the number. For example, to round a given number to the nearest tenth, we look one digit to the right of the tenths place (the hundredths place) and if it is greater than or equal to 5, we add one to the tenths place and remove all the digits to the right, otherwise we leave the tenths place as it is and remove all the digits to the right.

✓ Example 1.5.1

- a. 234.45 rounded to the nearest ten is 230.
- b. 45.6 rounded to the nearest ones (whole number) is 46.
- c. 34.555 rounded to the nearest tenth is 34.6.
- d. 34.54 rounded to the nearest tenth is 34.5.
- e. 34.95 rounded to the nearest tenth is 35.0.
- f. 34.554 rounded to the nearest hundredth is 34.55.
- g. 56.7874778 rounded to the nearest ten-thousandth is 56.7875

Adding and Subtracting Decimal Numbers

To add decimals, we line up the decimal points, and wherever there is a missing digit, we fill it in with a zero. For example, add 45.23 and 2.3:

$$\begin{array}{r} 45.23 \\ + 2.30 \\ \hline 47.53 \end{array}$$

Subtracting is similar. To subtract 45.23 from 2.3 we first note that the answer should be negative and proceed to subtract 2.3 from 45.23:

$$\begin{array}{r} 45.23 \\ - 2.30 \\ \hline 42.93 \end{array}$$

So, the answer of $2.3 - 45.23 = -42.93$

✓ Example 3.2

- a. $2.4 + 32.032 = 34.432$
- b. $3.44 + 12.035 = 15.475$
- c. $34.3 - 0.05 = 34.25$
- d. $6.3 - 9.72 = -3.42$

Multiplying and Dividing Decimal Numbers

Multiplying and Dividing Decimal Numbers by 10, 100, 1000, . . .

We first look at the special multiplication of decimals by 10, 100, 1000, . . .

$$\begin{array}{ll} 12.415 \times 10 & = 124.15 \\ 12.415 \times 100 & = 1241.5 \\ 12.415 \times 1000 & = 12415 \end{array}$$

When we multiply by 10 we move the decimal point to the right one place (because 10 has one decimal place). Multiplying by 100 moves the decimal point two places (because 100 has two decimal places), etc.

$$12.415 \div 10 = 1.2415$$

$$12.415 \div 100 = 0.12415$$

$$12.415 \div 1000 = 0.012415$$

When we divide by 10 we move the decimal point to the left one place (because 10 has one decimal place). Dividing by 100 moves the decimal point to the left two places (because 100 has two decimal places), etc.

10^n notation

$$10 = 10^1$$

$$100 = 10 \times 10 = 10^2$$

$$1000 = 10 \times 10 \times 10 = 10^3$$

Notice that the exponent of 10 in 10^n notation reflects the number of zeros! So, $10000 = 10^4$ (4 zeros, exponent is 4) and $100,000 = 10^5, \dots$

Multiplying by 10^n

Multiplying a decimal number by 10^n moves the decimal place n spots to the right. For example:

$$5.435 \times 10 = 54.35$$

$$5.435 \times 100 = 543.5$$

$$5.435 \times 10000 = 54350$$

Multiplying Decimal Numbers

To multiply two decimal numbers, we multiply as if there is no decimal point, then place a decimal point as described in the next example.

Example 3.3

Multiply 5.4 by 1.21.

Solution

$$\begin{array}{r} 1 \ 2 \ 1 \\ \times 5 \ 4 \\ \hline 4 \ 8 \ 4 \\ 6 \ 0 \ 5 \ 0 \\ \hline 6 \ 5 \ 3 \ 4 \end{array}$$

Now, to write out the answer, we notice that there are two digits after the decimal point in the first number 1.21, and one digit after the decimal point in the second number 5.4. The product then should have 3 digits after the decimal point. So, $5.4 \times 1.21 = 6.534$

Example 3.4

Multiply 3.72 by 13.

Solution

$$\begin{array}{r} 3 \ 7 \ 2 \\ \times 1 \ 3 \\ \hline 1 \ 1 \ 1 \ 6 \\ 3 \ 7 \ 2 \ 0 \\ \hline 4 \ 8 \ 3 \ 6 \end{array}$$

Now, to write out the answer, we notice that there are two digits after the decimal point in 3.72 while 13 has no decimal part. The product then should have 2 digits after the decimal point: $3.72 \times 13 = 48.36$

Dividing Decimal Numbers

Dividing a decimal number is a lot like dividing a whole number, except you use the position of the decimal point in the dividend to determine the decimal places in the result.

✓ Example 3.5

- a. $6.5 \div 2$
- b. $55.318 \div 3.4$

Solution

a) Divide as usual:

If the divisor does not go into the dividend evenly, add zeros to the right of the last digit in the dividend and continue until either the remainder is 0, or a repeating pattern appears. Place the position of the decimal point in your answer directly above the decimal point in the dividend.

$$\begin{array}{r} 3.25 \\ 2 \overline{) 6.5} \\ \underline{-6} \\ 05 \\ \underline{-4} \\ 10 \\ \underline{-10} \\ 0 \end{array}$$

b)

If the divisor is not a whole number, move the decimal point in the divisor all the way to the right (to make it a whole number). Then move the decimal point in the dividend the same number of places.

In this example, move the decimal point one place to the right for the divisor from 3.4 to 34. Therefore, also move the decimal point one place to the right for the dividend, from 55.318 to 553.18.

$$\begin{array}{r} 16.27 \\ 34 \overline{) 553.18} \\ \underline{-34} \\ 213 \\ \underline{-204} \\ 91 \\ \underline{-68} \\ 238 \\ \underline{-238} \\ 0 \end{array}$$

Converting Decimals to Fractions

To convert a decimal to a fraction is as simple as recognizing the place of the right most digit.

Example: Note that in the number 2.45, the right most digit 5 is in the hundredths place so $2.45 = \frac{245}{100} = \frac{49}{20}$ or $2\frac{9}{20}$

✓ Example 3.6

Here are a few more examples:

- a. $1.2 = \frac{12}{10} = \frac{6}{5}$ or $1\frac{1}{5}$
- b. $0.0033 = \frac{33}{10,000}$
- c. $0.103 = \frac{103}{1000}$

Converting Fractions to Decimals

To convert a fraction to a decimal you simply perform [long division](#).

✓ Example 3.7

Convert the given fraction to a decimal:

- a. $\frac{4}{5} = 4 \div 5 = 0.8$
- b. $3\frac{4}{5} = 3 + 4 \div 5 = 3.8$
- c. $\frac{13}{2} = 6\frac{1}{2} = 6 + 1 \div 2 = 6.5$
- d. (round to the nearest tenth) $\frac{3}{7} = 3 \div 7 = 0.42857 \dots \approx 0.4$

Converting Decimals to Percents and Percents to Decimals

"Percent" comes from Latin and means per hundred. We use the sign % for Por example, if you know that 25% of the students speak Spanish fluently, it means that 25 of every 100 students speak fluent Spanish. Presented as fraction, it would be $\frac{25}{100}$ and as a decimal 0.25.

✓ Example 3.8

Convert the given percent to fraction then to a decimal:

- a. 17% is $\frac{17}{100} = 0.17$
- b. 31% is $\frac{31}{100} = 0.31$
- c. 23.44% is $\frac{23.44}{100} = 0.2344$

✓ Example 3.9

Convert the given decimal to a fraction then to percent:

- a. $0.55 = \frac{55}{100}$ which is 55%
- b. $8.09 = \frac{809}{100}$ which is 809%
- c. $98.08 = \frac{9808}{100}$ which is 9808%
- d. $0.5 = \frac{50}{100}$ which is 50%

? Exit Problem

Divide: $782.56 \div 3.2$

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1.6: Evaluating Expressions

If a family pays \$30 per phone line, and \$20 per 1GB data, per month to a phone service provider, we can write a mathematical expression to represent the cost that this family pays per month.

Let's use x to represent the number of phone lines that the family has, and y to represent the number of GB of data the family uses. Then $30x + 20y$ is a mathematical expression that represents the cost of the family phone services per month.

$30x + 20y$ is an **algebraic expression**. A mathematical expression that consists of variables, numbers and algebraic operations is called an algebraic expression.

Each algebraic expression can contain several terms. For example, the expression above contains two terms: $30x$ and $20y$. The numerical factor of each term is called a coefficient. The coefficients of the terms above are 30 and 20, respectively. When considering a variable term, we see that it is composed of a **numerical coefficient** and a **variable part**. In the term $30x$ the numerical coefficient is 30 and the variable part is x .

The value of an algebraic expression can vary. For example, the value of the expression above can vary depending on the number of phone lines and the number of GB of data the family uses.

If $x = 2$ (the family uses 2 lines), and $y = 3$ (the family uses 3GB of data), then the cost of the family phone services is $30 \cdot 2 + 20 \cdot 3 = \120 for this month.

If $x = 4$, and $y = 2$, then the cost of the family phone devices is $30 \cdot 4 + 20 \cdot 2 = \160 for this month.

Finding the value of the expressions when the variables are substituted by given values is called evaluating an algebraic expression.

✓ Example 4.1

The algebraic expression $5x^3y - 2y^2 - z + 4$, which we write using only addition as $5x^3y + (-2y^2) + (-z) + 4$, contains four terms: $5x^3y$, $-2y^2$, $-z$ and $+4$. The first three terms are **variable terms** and the 4 is the **constant term**. Notice that the coefficient of $-z$ is -1 , and we usually write $-z$ instead of $-1z$. In the same way, if the coefficient is 1, we usually omit it.

This process of finding the value of an algebraic expression for particular values of its variables is called **evaluating an expression**.

📌 Evaluating an expression

1. Replace each variable by the given numerical value.
2. Simplify the resulting expression. Be careful to follow the order of operations.

Helpful tip: When evaluating a variable expression containing a fraction bar, don't forget to work out the numerator and denominator separately (being careful to follow the order of operations as you do so); finally, divide the numerator by the denominator.

✓ Example 4.2

Evaluate if $a = 1$, $b = 2$, $c = 4$, and $d = -1$:

- $8b$
- a) $= 8(2)$
 $= 16$
- $a + c$
- b) $= 1 + 4$
 $= 5$
- $a - d$
- c) $= 1 - (-1)$
 $= 1 + 1$
 $= 2$

$$\begin{aligned} & 5ab \\ \text{d)} &= 5(1)(2) \\ &= 10 \end{aligned}$$

$$\begin{aligned} & \frac{d+a}{b} \\ \text{e)} &= \frac{(-1)+1}{2} \\ &= \frac{0}{2} \\ &= 0 \end{aligned}$$

$$\begin{aligned} & \frac{7b+c}{a-d} \\ \text{f)} &= \frac{7(2)+4}{1-(-1)} \\ &= \frac{14+4}{1+1} \\ &= \frac{18}{2} \\ &= 9 \end{aligned}$$

✓ Example 4.3

Evaluate if $a = -3$, $b = 5$, $c = -2$, and $d = 7$:

$$\begin{aligned} & 4c - 2b \\ \text{a)} &= 4(-2) - 2(5) \\ &= -8 - 10 \\ &= (-8) + (-10) \\ &= -18 \end{aligned}$$

$$\begin{aligned} & b^2 + b \\ \text{b)} &= 5^2 + 5 \\ &= 25 + 5 \\ &= 30 \end{aligned}$$

$$\begin{aligned} & 3c^2 \\ \text{c)} &= 3(-2)^2 \\ &= 3(4) \\ &= 12 \end{aligned}$$

$$\begin{aligned} & (c+a)(c^2-ac+a^2) \\ \text{d)} &= ((-2)+(-3))((-2)^2-(-3)(-2)+(-3)^2) \\ &= (-5)(4-(-3)(-2)+9) \\ &= (-5)(4-6+9) \\ &= (-5)(7) \\ &= -35 \end{aligned}$$

$$\begin{aligned} & 4b + 5d - \frac{c}{a} \\ \text{e)} &= 4(5) + 5(7) - \frac{(-2)}{(-3)} \\ &= 4(5) + 5(7) - \frac{2}{3} \\ &= 20 + 35 - \frac{2}{3} \\ &= \frac{60}{3} + \frac{105}{3} - \frac{2}{3} \\ &= \frac{60+105-2}{3} \\ &= \frac{163}{3} \\ &= 54\frac{1}{3} \end{aligned}$$

$$\text{f)} -d^2 = -(7)^2 = -49$$

Application

Here are some applications that come from geometry.

✓ Example 4.4

a) Find the perimeter of a rectangle with length 33cm and width 17cm.

$$\begin{aligned}P &= 2l + 2w \\&= 2(33\text{cm}) + 2(17\text{cm}) \\&= 66\text{cm} + 34\text{cm} \\&= 100\text{cm}\end{aligned}$$

b) Find the area of a rectangle that is 12cm long and 16cm wide.

$$\begin{aligned}A &= l \cdot w \\&= 12\text{cm} \cdot 16\text{cm} \\&= 192\text{cm}^2\end{aligned}$$

c) Find the area of a triangle with height of 20in and a base of 30in.

$$\begin{aligned}A &= \frac{1}{2}b \cdot h \\&= \frac{1}{2}20\text{in} \cdot 30\text{in} \\&= 300\text{in}^2\end{aligned}$$

d) Find the area of a circle with radius 7cm, round your answer to the nearest tenth.

$$\begin{aligned}A &= \pi r^2 \\&= \pi(7\text{cm})^2 \\&\approx 3.14 \cdot 49\text{cm}^2 \\&= 153.86\text{cm}^2 \\&\approx 153.9\text{cm}^2\end{aligned}$$

? Exit Problem

Evaluate if $a = -2$: $-3a^2 - 4a - 16$

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1.7: Properties of Exponents

Multiplication Properties

Recall from chapter 1 that $5^3 = 5 \cdot 5 \cdot 5$. In the same way we have

$$x^3 = x \cdot x \cdot x$$

✓ Example 5.1

Perform the given operations:

- $x^3 \cdot x^5 = \underbrace{x \cdot x \cdot x}_{=x^3} \cdot \underbrace{x \cdot x \cdot x \cdot x \cdot x}_{=x^5} = x^8$
- $(x^3)^4 = x^3 \cdot x^3 \cdot x^3 \cdot x^3 = \underbrace{x \cdot x \cdot x}_{=x^3} \cdot \underbrace{x \cdot x \cdot x}_{=x^3} \cdot \underbrace{x \cdot x \cdot x}_{=x^3} \cdot \underbrace{x \cdot x \cdot x}_{=x^3} = x^{12}$
- $(x \cdot y)^4 = \underbrace{x \cdot y}_{=xy} \cdot \underbrace{x \cdot y}_{=xy} \cdot \underbrace{x \cdot y}_{=xy} \cdot \underbrace{x \cdot y}_{=xy} = x^4 y^4$

We can summarize these examples into the following useful rules:

✚ Multiplication and Exponentiation Rules

For any integers n and m

- $x^n x^m = x^{n+m}$
- $(x^n)^m = x^{nm}$
- $(xy)^n = x^n y^n$

✓ Example 5.2

Perform the given operation using the multiplication properties of exponents and write your answer in simplest form:

- $b^2 \cdot b^3 = b^{2+3} = b^5$ (recall the meaning of exponents)
- $x^8 x^7 = x^{8+7} = x^{15}$ (note that juxtaposition indicates multiplication)
- $a^8 a^9 a^{14} = a^{8+9+14} = a^{31}$
- $4x^4 \cdot 7x^6 = (4 \cdot 7)(x^4 \cdot x^6) = 28x^{10}$
- $5xy^3 \cdot 6y = (5 \cdot 6)(x)(y^3 \cdot y) = 30xy^4$
- $(5x^4 y^2)(2x^7 y^3) = 10x^{4+7} y^{2+3} = 10x^{11} y^5$
- $(2x)^3 = 2^3 x^3 = 8x^3$
- $(-4a^2 b^5)^3 = (-4)^3 a^{2 \cdot 3} b^{5 \cdot 3} = -64a^6 b^{15}$
- $(-5r^3 s^2)^2 \cdot (2r^4 s^3)^3 \cdot (-r^2 s^2) = (-5)^2 r^{3 \cdot 2} s^{2 \cdot 2} \cdot 2^3 r^{4 \cdot 3} s^{3 \cdot 3} \cdot (-1)r^2 s^2$
 $= 25r^6 s^4 \cdot 8r^{12} s^9 \cdot (-1)r^2 s^2$
 $= -200r^{20} s^{13}$

Division Properties

✓ Example 5.3

- $\frac{x^5}{x^3} = \frac{x \cdot x \cdot x \cdot x \cdot x}{x \cdot x \cdot x} = \frac{x \cdot x}{1} = \frac{x^2}{1} = x^2$
- $\frac{x^5}{x^3} = \frac{x \cdot x \cdot x \cdot x \cdot x}{x \cdot x \cdot x} = \frac{x \cdot x}{x^2} = \frac{x^2}{x^2}$
- $\left(\frac{x}{y}\right)^5 = \frac{x}{y} \cdot \frac{x}{y} \cdot \frac{x}{y} \cdot \frac{x}{y} \cdot \frac{x}{y} = \frac{x \cdot x \cdot x \cdot x \cdot x}{y \cdot y \cdot y \cdot y \cdot y} = \frac{x^5}{y^5}$

We can summarize these examples into the following useful rules:

Division Rules

For any integers n and m

1. $\frac{x^n}{x^m} = x^{n-m}$
2. $\left(\frac{x}{y}\right)^n = \frac{x^n}{y^n}$

✓ Example 5.4

Perform the given operation using the division properties of exponents and state your answer in simplest form:

- a. $\frac{b^8}{b^7} = b^{8-7} = b^{8+(-7)} = b^1 = b$
- b. $\frac{x^{12}y^2}{x^8y^2} = x^{12-8}y^{2-2} = x^{12+(-8)}y^{2+(-2)} = x^4y^0 = x^4y$
- c. $\frac{8x^6y^2}{6x^5y^7} = \frac{4x^{6-5}}{3y^{7-2}} = \frac{4x^1}{3y^5} = \frac{4x}{3y^5}$
- d. $\frac{(m^2)^3(n^4)^5}{(m^3)^3} = \frac{m^6n^{20}}{m^9} = \frac{n^{20}}{m^3}$
- e. $\left(\frac{3a^2b^4}{9c^3}\right)^2 = \left(\frac{a^2b^4}{3c^3}\right)^2 = \frac{a^{2 \cdot 2}b^{4 \cdot 2}}{3^2c^{3 \cdot 2}} = \frac{a^4b^8}{9c^6}$ Another way to simplify this correctly is this: $\left(\frac{3a^2b^4}{9c^3}\right)^2 = \frac{3^2a^{2 \cdot 2}b^{4 \cdot 2}}{9^2c^{3 \cdot 2}} = \frac{9a^4b^8}{81c^6} = \frac{a^4b^8}{9c^6}$

Zero Exponent

Recall from chapter 1 that $(-7)^0 = 1$ and $8^0 = 1$. In the same way we have $x^0 = 1$

Zero Exponent

$$a^0 = 1$$

✓ Example 5.5

Evaluate

- a. $15^0 = 1$
- b. $(-15)^0 = 1$
- c. $-15^0 = -1$
- d. $(15x)^0 = 1$
- e. $15x^0 = 15 \cdot 1 = 15$

Negative Exponents

Negative Exponents

For any integer n

1. $a^{-n} = \frac{1}{a^n}$
2. $\frac{1}{a^{-n}} = a^n$
3. $\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n$

Note 5.6. The negative exponent rules can be used to switch terms from numerator to denominator or vice versa, and is useful to write expressions using positive exponents only.

✓ Example 5.7

$$\begin{aligned}\text{a. } \frac{x^{-3}y^2}{z^4} &= \frac{y^2}{x^3z^4} \\ \text{b. } \frac{x^4}{y^{-5}z^2} &= \frac{x^4y^5}{z^2}\end{aligned}$$

✓ Example 5.8

Perform the given operation and write your answer using positive exponents only

$$\begin{aligned}\text{a. } \frac{x^4y^2}{x^3y^{-3}} &= x^{4-3}y^{2-(-3)} = x^{4+(-3)}y^{2+3} = xy^5 \\ \text{b. } (x^2y) \cdot (xy^{-4}) &= x^{2+1}y^{1+(-4)} = x^3y^{-3} = \frac{x^3}{y^3}\end{aligned}$$

? Exit Problem

Simplify: $\left(\frac{8y}{3x^3}\right)^3$

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1.8: Scientific Notation

To write 1 trillion (1 followed by 12 zeros) or 1 googol (1 followed by 100 zeroes) takes a lot of space and time. There is a mathematical scientific notation which is very useful for writing very big and very small numbers.

✓ Example 1.8.1

Large numbers in scientific notation:

- 1 trillion is written as 1×10^{12} in scientific notation.
- 4 trillion is written as 4×10^{12} in scientific notation.
- 1 googol is written as 1×10^{100} in scientific notation.

✓ Example 1.8.2

- 0.00000547 is written as 5.47×10^{-6} in scientific notation.
- 0.00031 is written as 3.1×10^{-4} in scientific notation.

The number 45,600,000 is a large number, and, basically is $4.56 \times 10,000,000$. So, it can be written as 4.56×10^7

Similarly, if we consider the number 0.00006772. This is a small number which is $6.772 \times \frac{1}{100000}$. That is, it can be written as 6.772×10^{-5} .

The numbers 4.56×10^7 and 6.772×10^{-5} are said to be written in scientific notation because the number before the power of 10 is greater than (or equal) to 1 and less than 10, and the decimal number is followed by multiplication by a power of 10.

📌 Standard Form and Scientific Notation

<u>Standard form</u>	<u>Scientific notation</u>
34,500,000,000	3.45×10^{10}
0.00000000889	8.89×10^{-9}

Recall from chapter 3 how multiplying or dividing a decimal number by 10, 100, 100, ... affects the position of the decimal point.

✓ Example 1.8.3

The given numbers are not in scientific notation. Modify each so that your answer is in scientific notation:

- $1500 = 1.5 \times 10^3$
- $225000 = 2.25 \times 10^5$
- $0.0155 = 1.55 \times 10^{-2}$
- $0.00000094 = 9.4 \times 10^{-7}$

✓ Example 1.8.4

The given numbers are not in scientific notation (look at the decimal number and see that it is either less than 1 or greater than 10). Modify each so that your answer is in scientific notation:

- $56.7 \times 10^8 = 5.67 \times 10^9$
- $88.9 \times 10^{-7} = 8.89 \times 10^{-6}$
- $0.55 \times 10^9 = 5.5 \times 10^8$
- $0.88 \times 10^{-4} = 8.8 \times 10^{-5}$

Helpful tip: Note that the given numbers were not in scientific notation because the decimal number was either greater than 10 or less than 1. To modify the decimal and rewrite the given number in scientific notation, we either increase its size, and thus we must decrease the size of the exponent, or, we decrease its size, and thus, we must increase the size of the exponent.

Multiplication and Division using Scientific Notation

By grouping the decimal numbers together, and the power of 10 terms together, it becomes easy to multiply and divide numbers in scientific notation. First, we need to recall the properties of exponents (we only need base 10 for this section):

Properties of Exponents (for base 10)

1. Product Property $10^m \cdot 10^n = 10^{m+n}$

Examples: $10^2 \cdot 10^5 = 10^{2+5} = 10^7$ and $10^{-9} \cdot 10^3 = 10^{-9+3} = 10^{-6}$

2. Quotient Property $\frac{10^m}{10^n} = 10^{m-n}$

Examples: $\frac{10^5}{10^3} = 10^{5-3} = 10^2$ and $\frac{10^5}{10^{-4}} = 10^{5-(-4)} = 10^{5+4} = 10^9$

✓ Example 1.8.5

Perform the given operation:

a. $(2.3 \times 10^8)(3 \times 10^{-4}) = (2.3 \cdot 3) \times (10^8 \cdot 10^{-4}) = (2.3 \cdot 3) \times 10^{8+(-4)} = 6.9 \times 10^4$

b. $\frac{6.4 \times 10^{-9}}{3.2 \times 10^{-5}} = \frac{6.4}{3.2} \times \frac{10^{-9}}{10^{-5}} = \frac{6.4}{3.2} \times 10^{-9-(-5)} = 2 \times 10^{-4}$

Helpful tip: Notice how when we multiplied, we added the exponents, and when we divided, we subtracted the exponent in the denominator from the exponent in the numerator. We simply followed the exponent rules.

✓ Example 1.8.6

Perform the given operation and write your answer in scientific notation:

a. $(6.2 \times 10^8)(3.0 \times 10^7) = 6.2 \cdot 3 \times 10^{8+7} = 18.6 \times 10^{15} = 1.86 \times 10^{16}$

b. $\frac{4 \times 10^5}{8 \times 10^{-3}} = \frac{4}{8} \times 10^{5-(-3)} = 0.5 \times 10^8 = 5.0 \times 10^7$

Note: Writing 5.0×10^7 is the same as writing 5×10^7 . They are interchangeable.

✓ Example 1.8.7

Perform the given operation and write your answer in scientific notation:

$$\frac{(2.1 \times 10^3)(3.2 \times 10^{-8})}{(2 \times 10^4)(3 \times 10^9)} = \frac{6.72 \times 10^{-5}}{6 \times 10^{13}} = 1.12 \times 10^{-18}$$

✓ Example 1.8.8

The debt of a nation is 7 trillion dollars, and there are 300 thousand inhabitants. If the debt was distributed evenly among all the inhabitants, how much would each person have to pay to pay off the debt?

To answer this question we have to divide: $\frac{7 \text{ trillion dollars}}{300 \text{ million people}}$.

Since we are dealing with large numbers, we change each to scientific notation and perform the division to find the amount to be paid per person:

$$\frac{7 \times 10^{15}}{3 \times 10^{11}} \text{ dollars per person} = \frac{7}{3} \times 10^4 \text{ dollars per person} \approx 2.3333 \times 10^4 \text{ dollars per person.}$$

In standard form, the amount is \$23,333 per person.

? Exit Problem

Compute and write the answer in scientific notation:

$$\frac{(6.2 \times 10^2)(1.5 \times 10^{-4})}{3.1 \times 10^{-9}}$$

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1.9: Polynomials

A polynomial is a sum of monomials. So, expressions like:

$$\begin{aligned}x^2 + 3x + 7 \\ -2x^3 + 4x^2 - 5x + 2 \\ x + x^2 \\ -4x^3 \\ x^2y + \frac{xyz^2}{6} - 8y^2z^2\end{aligned}$$

are examples of polynomials. However, the following are *not* polynomials:

$$\begin{aligned}\frac{x^2 + 3x + 4}{x + 5} \\ \frac{2x^2yz^3}{-xy^2} \\ -4x\sqrt{6x}\end{aligned}$$

The **degree** of a polynomial is the highest power of the variable(s) that has a non-zero coefficient.

✓ Example 1.9.1

The degree of $-2x^3 + 4x^2 - 5x + 2$ is 3.

✓ Example 1.9.2

Determine whether the given expression is a polynomial and if so, find its degree.

a) $5x^3 + 4x^2 - 3x + 7$

This is a polynomial of degree 3.

b) $\frac{2}{x^2 + 3x - 4}$

This is not a polynomial.

c) $2x^3 + 5x^4 + 3x - 8$

This is a polynomial of degree 4.

d) $6.2 \times 10^{-5}x^8$

This is a polynomial of degree 8.

A polynomial with one term is called a **monomial**. For example, $2a^5$ and $-3xy^2$ are monomials. A polynomial with two terms is called a **binomial**. $5x^2 + 3x$ is an example of a binomial. A polynomial with three terms is called a trinomial. $3x^2 + 5x - 1$ is a **trinomial**. It has three terms: $3x^2$, $5x$ and -1

Just as we did in chapter 4 when evaluating expressions, we can evaluate polynomials as well.

✓ Example 1.9.3

Evaluate the given polynomial at the given value of the variable(s):

a. $3x + 7$ when $x = 2$:

$$3(2) + 7 = 6 + 7 = 13$$

b. $x^2 + 3x + 2$ when $x = 5$:

$$(5)^2 + 3(5) + 2 = 25 + 15 + 2 = 42$$

c. $2x^3 + 4x^2 - 3x$ when $x = -2$:

$$2(-2)^3 + 4(-2)^2 - 3(-2) = 2(-8) + 4(4) - (-6) = -16 + 16 + 6 = 6$$

d. $-4x^7 - 3x^4$ when $x = -1$:

$$-4(-1)^7 - 3(-1)^4 = -4(-1) - 3(1) = 4 - 3 = 1$$

e. $2x^2y - 5x^3y^2$ when $x = -3$ and $y = 2$:

$$\begin{aligned} 2(-3)^2(2) - 5(-3)^3(2)^2 &= 2(9)(2) - 5(-27)(4) \\ &= 18(2) + 135(4) \\ &= 36 + 540 \\ &= 576 \end{aligned}$$

Function Notation

A specific kind of notation, called **function notation**, can be used to represent polynomials. This notation uses a letter (the name of the function) and the variable at hand (for example, x .)

✓ Example 1.9.4

- a. $f(x) = 3x + 7$ is representing the polynomial $3x + 7$ as a function called f . The x in $f(x)$ is to indicate that the variable in the polynomial is x .
- b. $g(x) = x^2 + 3x + 2$ is representing the polynomial $x^2 + 3x + 2$ as a function called g . Again, the x in $g(x)$ is to indicate that the variable in the polynomial is x .
- c. $f(x, y) = 2x^2y - 5x^3y^2$ is representing the polynomial $2x^2y - 5x^3y^2$ as a function called f . The x and y in $f(x, y)$ are to indicate that the variables in the polynomial are x and y .

You will learn about functions and function notation in a pre-calculus class, but, here we use the notation because it facilitates asking to evaluate a polynomial at a given value of the variable(s), as we saw in Example 7.3.

So, for example, find $f(2)$ when $f(x) = 3x + 7$ is asking to evaluate the polynomial $3x + 7$ when $x = 2$. So, $f(2) = 3 \cdot 2 + 7 = 13$.

✓ Example 1.9.5

$f(x) = x^2 - 1$. Find $f(-3)$

Solution

$$f(-3) = (-3)^2 - 1 = 9 - 1 = 8$$

✓ Example 1.9.5

$g(x) = -3x^3 + 5$. Find $g(-2)$.

Solution

$$g(-2) = -3 \cdot (-2)^3 + 5 = -3 \cdot (-8) + 5 = 24 + 5 = 29$$

? Exit Problem

Evaluate $f(-1)$ for the function $f(x) = -2x^3 + 3x^2 - x$

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1.10: Adding and Subtracting Polynomial Expressions

Polynomials can be **added**, **subtracted**, **multiplied** and **divided**. When **adding** or **subtracting**, we can only combine terms that are **like terms**.

What Are Like Terms?

Consider the expression

$$5x^4y^2 + 6x^3y - 7y^2x^4$$

First, we rewrite the given expression using addition only:

$$5x^4y^2 + 6x^3y + (-7y^2x^4)$$

Here the terms are: $5x^4y^2$, $6x^3y$, and $-7y^2x^4$. The terms are added to get the expression.

Two terms are either “unlike” or “like”.

Like Terms

Like terms are terms that have the same exponents on the same variables.

For example, in the above expression $5x^4y^2$ and $-7y^2x^4$ are like terms since y has the same exponent (2) and the x has the same exponent (4). On the other hand, $5x^4y^2$ and $6x^3y^2$ are unlike terms since x^4 appears in the first term but x^3 appears in the second.

✓ Example 8.1

Consider the expression:

$$-2x^5y^2 - 5x^4y^2 + 6x^5y^2$$

Solution

The like terms are: $-2x^5y^2$ and $6x^5y^2$.

Adding or Subtracting Like Terms

Adding or Subtracting Like Terms

We can only add or subtract like terms, and, we do so by adding or subtracting their coefficients.

✓ Example 8.2

$-2x^5y^2 + 6x^5y^2 = 4x^5y^2$ so that our expression in the previous example can be simplified:

$$-2x^5y^2 - 5x^4y^2 + 6x^5y^2 = 4x^5y^2 - 5x^4y^2$$

Now, let's add and subtract polynomials.

✓ Example 8.3

Add or subtract the polynomials.

$$\begin{aligned} \text{a) } (3x^2 + 5x + 6) + (4x^2 + 3x - 8) &= 3x^2 + 5x + 6 + 4x^2 + 3x - 8 \\ &= 7x^2 + 8x - 2 \end{aligned}$$

$$\begin{aligned} \text{b) } (x^4 - 2x^3 + 6x) + (5x^3 + 2x^2 + 9x + 4) &= x^4 - 2x^3 + 6x + 5x^3 + 2x^2 + 9x + 4 \\ &= x^4 + 3x^3 + 2x^2 + 15x + 4 \end{aligned}$$

$$\begin{aligned} \text{c) } (2ab^2 - 3a^2 - 7ab) + (-a^2 - 5a^2b) &= 2ab^2 - 3a^2 - 7ab - a^2 - 5a^2b \\ &= 2ab^2 - 4a^2 - 7ab - 5a^2b \end{aligned}$$

Note: Above, the only like terms that can be combined are $-3a^2$ and $-a^2$. The remaining terms cannot be combined any further.

$$\begin{aligned} \text{d) } (3x^2 + 5x + 6) - (4x^2 + 3x - 8) &= 3x^2 + 5x + 6 - 4x^2 - 3x + 8 \\ &= -x^2 + 2x + 14 \end{aligned}$$

Note: Removing the parenthesis when subtracting changes the sign for all the terms of the polynomial that is being subtracted.

$$\begin{aligned} \text{e) } (5m + 3n) - (7m + 2n) &= 5m + 3n - 7m - 2n \\ &= -2m + n \end{aligned}$$

f) Subtract $4a^2 - 5a$ from $6a + 4$.

$$\begin{aligned} (6a + 4) - (4a^2 - 5a) &= 6a + 4 - 4a^2 + 5a \\ &= -4a^2 + 11a + 4 \end{aligned}$$

Note: Subtracting $4a^2 - 5a$ from $6a + 4$ requires to write $6a + 4$ first, and then to subtract $4a^2 - 5a$ from it. Reversing the order would yield a wrong answer!

g) Subtract $-3q + 4pq - 2p$ from $-9p$.

$$\begin{aligned} -9p - (-3q + 4pq - 2p) &= -9p + 3q - 4pq + 2p \\ &= -7p + 3q - 4pq \end{aligned}$$

? Exit Problem

Simplify: $(9m^2n - 15mn^2) - (3mn^2 + 2m^2n)$

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1.11: Multiplying Polynomial Expressions

In this chapter, we multiply polynomials. For the **multiplication** of a monomial by a polynomial, we need to distribute the monomial to multiply each term of the polynomial.

Theorem: Distributive Law

$$a(b + c) = ab + ac \quad (1.11.1)$$

By distributing factors, we can multiply a monomial by a polynomial

✓ Example 1.11.1

Multiply the terms and simplify.

a) $3(4x^2 + 5x) = 12x^2 + 15x$

b) $2p(4p - 7q) = 8p^2 - 14pq$

c) $-3a(a^2 - 4a + 5) = -3a^3 + 12a^2 - 15a$

d) $6x + 7y + 4(3x + 2y) = 6x + 7y + 12x + 8y$
 $= 18x + 15y$

e) $2(3x^2 + 5) + 3x(4x - 2) = 6x^2 + 10 + 12x^2 - 6x$
 $= 18x^2 - 6x + 10$

f) $y^2(y + 3) - 2y(-y + 5) - 4(2y^2 + 12y + 1) = y^3 + 3y^2 + 2y^2 - 10y - 8y^2 - 48y - 4$
 $= y^3 - 3y^2 - 58y - 4$

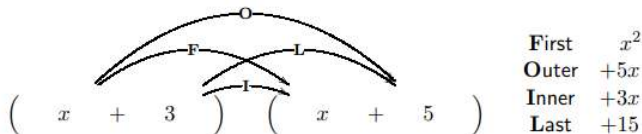
g) $-4ab(-a - 3b) + 2a(8b^2 - 7ab) = 4a^2b + 12ab^2 + 16ab^2 - 14a^2b$
 $= -10a^2b + 28ab^2$

When multiplying general polynomials, we need to have a more general distributive law. We show how this is done for the product of two binomials.

Rule 1.11.2: FOIL

To multiply binomials, we need to multiply each term of the first polynomial with each term of the second polynomial. This procedure is known under the name **FOIL**, which stands for **F**irst, **O**uter, **I**nnner, **L**ast.

For example, when multiplying $(x + 3)$ with $(x + 5)$, we multiply the terms



The products of the first terms ($x \cdot x = x^2$), the outer terms ($x \cdot 5 = 5x$), the inner terms ($3 \cdot x = 3x$), and the last terms ($3 \cdot 5 = 15$) are added to give the result. Therefore, applying FOIL and combining like terms, we obtain:

$$(x + 3)(x + 5) = x^2 + 5x + 3x + 15 = x^2 + 8x + 15$$

✓ Example 9.3

Multiply and simplify.

a) $(x + 2)(x - 7) = x^2 - 7x + 2x - 14$
 $= x^2 - 5x - 14$

b) $(2a + 3b)(4a + 5b) = 8a^2 + 10ab + 12ab + 15b^2$
 $= 8a^2 + 22ab + 15b^2$

$$\begin{aligned}
 (m-4n)^2 &= (m-4n)(m-4n) \\
 \text{c)} \quad &= m^2 - 4mn - 4mn + 16n^2 \\
 &= m^2 - 8mn + 16n^2 \\
 -5p(3p+2q)^2 &= -5p(3p+2q)(3p+2q) \\
 \text{d)} \quad &= -5p(9p^2 + 6pq + 6pq + 4q^2) \\
 &= -5p(9p^2 + 12pq + 4q^2) \\
 &= -45p^3 - 60p^2q - 20pq^2
 \end{aligned}$$

Above we have chosen to first evaluate the square $(3p+2q)(3p+2q)$. Alternatively, we could also have first multiplied $-5p(3p+2q)$, which also gives the correct result:

$$\begin{aligned}
 -5p(3p+2q)^2 &= -5p(3p+2q)(3p+2q) \\
 &= (-15p^2 - 10pq)(3p+2q) \\
 &= -45p^3 - 30p^2q - 30p^2q - 20pq^2 \\
 &= -45p^3 - 60p^2q - 20pq^2
 \end{aligned}$$

$$\begin{aligned}
 (r+s)^2 - (r+s)(r-s) &= (r+s)(r+s) - (r+s)(r-s) \\
 &= (r^2 + rs + rs + s^2) - (r^2 - rs + rs - s^2) \\
 \text{e)} \quad &= (r^2 + 2rs + s^2) - (r^2 - s^2) \\
 &= r^2 + 2rs + s^2 - r^2 + s^2 \\
 &= 2rs + 2s^2 \\
 2xy - 3(5x+y)(5x-y) + 4y(3x+2y) &= 2xy - 3(25x^2 - 5xy + 5xy - y^2) + 12xy + 8y^2 \\
 \text{f)} \quad &= 2xy - 3(25x^2 - y^2) + 12xy + 8y^2 \\
 &= 2xy - 75x^2 + 3y^2 + 12xy + 8y^2 \\
 &= -75x^2 + 14xy + 11y^2 \\
 (2x-1)(x^2-4x+6) &= (2x+(-1))(x^2-4x+6) \\
 &= 2x(x^2-4x+6) + (-1)(x^2-4x+6) \\
 \text{g)} \quad &= 2x \cdot x^2 + 2x \cdot (-4x) + 2x \cdot 6 + (-1) \cdot x^2 + (-1) \cdot (-4x) + (-1) \cdot 6 \\
 &= 2x^3 - 8x^2 + 12x - x^2 + 4x - 6 \\
 &= 2x^3 - 9x^2 + 16x - 6
 \end{aligned}$$

? Exit Problem

Multiply: $(5x-2)(x^2-3x-7)$

Simplify: $-3a^3b^2 + 4a^2(a+2ab^2) - 7a^3$

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1.12: Dividing Polynomials

In the previous chapter, we added, subtracted, and multiplied polynomials. Now, what remains is dividing polynomials. We will only consider division of a polynomial by a monomial. The division of a monomial by a monomial was already considered in Chapter 5, which we now recall.

Rule 1.12.1

We recall the rules for dividing variables.

$$\frac{x^n}{x^m} = x^{n-m} \quad (1.12.1)$$

for any integers n and m

Example 1.12.2

Simplify.

$$\begin{aligned} \text{a. } \frac{27x^3y^5}{3x^2y^3} &= \frac{27x^{3-2}y^{5-3}}{3} = 9x^1y^2 = 9xy^2 \\ \text{b. } \frac{-56a^8b^6c^4}{-7a^5bc^4} &= \frac{-56a^{8-5}b^{6-1}c^{4-4}}{-7} = 8a^3b^5c^0 = 8a^3b^5 \end{aligned}$$

Of course, when the power of a variable is higher in the denominator than in the numerator, then those variables will remain in the denominator, just as we did in chapter 5.

Example 1.12.3

Simplify.

$$\begin{aligned} \text{a. } \frac{42p^7q^4}{-3p^3q^2} &= \frac{42}{-3}q^{4-2}p^{7-4} = \frac{14}{-1}q^2p^3 = -14q^2p^3 \\ \text{b. } \frac{24r^4s^9t^5}{20rs^6t^2} &= \frac{24}{20}r^{4-1}s^{9-6}t^{5-2} = \frac{6}{5}r^3s^3t^3 \end{aligned}$$

We now study how a polynomial can be divided by a monomial. Recall the usual rule for adding fractions with common denominator.

Same Denominator Fractions

Fractions with common denominator can be added (or subtracted) by adding (or subtracting) the numerators:

- Add:

$$\frac{a}{c} + \frac{b}{c} = \frac{a+b}{c} \quad (1.12.2)$$

- Subtract:

$$\frac{a}{c} - \frac{b}{c} = \frac{a-b}{c} \quad (1.12.3)$$

Reversing the above rule helps us to divide a polynomial by a monomial.

Example 1.12.4

Simplify as much as possible.

$$\begin{aligned} \text{a) } \frac{6x+15}{3} &= \frac{6x}{3} + \frac{15}{3} = 2x+5 \\ \text{b) } \frac{14x^3-8x^2}{2x} &= \frac{14x^3}{2x} - \frac{8x^2}{2x} = 7x^2-4x \end{aligned}$$

$$\begin{aligned}
 \text{c) } \frac{14y^6 - 28y^5 + 21y^3}{-7y^2} &= \frac{14y^6}{-7y^2} - \frac{28y^5}{-7y^2} + \frac{21y^3}{-7y^2} \\
 &= -2y^4 - (-4y^3) - 3y \\
 &= -2y^4 + 4y^3 - 3y \\
 \text{d) } \frac{a^2b^4 - 4ab^3 - 2a^4b^2}{ab^2} &= \frac{a^2b^4}{ab^2} - \frac{4ab^3}{ab^2} - \frac{2a^4b^2}{ab^2} \\
 &= ab^2 - 4b - 2a^3 \\
 \text{e) } \frac{-6r^5t^4 + 30r^4s^2t^5 - 42r^3s^2t^3}{-6rt^3} &= \frac{-6r^5t^4}{-6rt^3} + \frac{30r^4s^2t^5}{-6rt^3} - \frac{42r^3s^2t^3}{-6rt^3} \\
 &= r^4t - 5r^3s^2t^2 - (-7r^2s^2) \\
 &= r^4t - 5r^3s^2t^2 + 7r^2s^2
 \end{aligned}$$

? Exit Problem

Simplify: $\frac{27x^2y - 3xy + 15xy^2}{-3xy}$

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1.13: Simplifying Square Roots

Finding a square root of a number is the inverse operation of squaring that number. Remember, the square of a number is that number times itself. For example,

$$5^2 = 5 \cdot 5 = 25$$

and

$$(-5)^2 = (-5) \cdot (-5) = 25.$$

The square root of a number n , written as $\sqrt{n}$, is the **positive number** that gives n when multiplied by itself. For example, $\sqrt{25} = 5$ and not -5 because 5 is the **positive** number that multiplied by itself gives 25 .

The perfect squares are the squares of whole numbers:

- $1 = 1^2$
- $4 = 2^2$
- $9 = 3^2$
- $16 = 4^2$
- $25 = 5^2$
- $36 = 6^2$
- $49 = 7^2$
- $64 = 8^2$
- $81 = 9^2$
- $100 = 10^2, \dots$

and finding their square roots is straightforward. So

$$\begin{aligned}\sqrt{16} &= \sqrt{4^2} \\ &= 4 \\ \sqrt{100} &= \sqrt{10^2} \\ &= 10.\end{aligned}$$

What about $\sqrt{50}$? Can you think of a number you multiply by itself and the answer is 50 ? The only thing we can do is simplify the square root. We say that a square root is **simplified** if it has **no perfect square factors**.

So, to simplify $\sqrt{50}$ we first write 50 into its factor and look for perfect squares.

$$50 = 25 \cdot 2 = 5^2 \cdot 2.$$

Then,

$$\begin{aligned}\sqrt{50} &= \sqrt{25 \cdot 2} \\ &= \sqrt{5^2} \cdot \sqrt{2} \\ &= 5 \cdot \sqrt{2}.\end{aligned}$$

The justification for separating $\sqrt{5^2}$ and $\sqrt{2}$ is the fact that the square root of a **product** is equal to the product of the square root of each factor:

Product Rule for Radicals

$$\sqrt{a \cdot b} = \sqrt{a} \cdot \sqrt{b} \quad (1.13.1)$$

Example 11.1

Simplify each of the following radical expressions:

$$\begin{aligned}
 \text{a. } \sqrt{24} &= \sqrt{4 \cdot 6} = \sqrt{2^2 \cdot 6} \\
 &= \sqrt{2^2} \cdot \sqrt{6} = 2 \cdot \sqrt{6} \\
 \text{b. } \sqrt{108} &= \sqrt{36 \cdot 3} = \sqrt{6^2 \cdot 3} = \sqrt{6^2} \cdot \sqrt{3} = 6 \cdot \sqrt{3} \\
 2 \cdot \sqrt{80} &= 2 \cdot \sqrt{16 \cdot 5} = 2 \cdot \sqrt{4^2 \cdot 5} \\
 \text{c. } &= 2 \cdot \sqrt{4^2} \cdot \sqrt{5} = 2 \cdot 4 \cdot \sqrt{5} \\
 &= 8 \cdot \sqrt{5}
 \end{aligned}$$

How to combine "like" square roots

We can combine "like" square roots the same way we combined "like terms" in [chapter 8](#).

Like Square Roots

Two (or more) square roots are "like" if they have the same quantity under the root.

Note: Always simplify the square root if possible before identifying "like" roots.

✓ Example 11.2

Like square roots:

- $\sqrt{3}$ and $-6\sqrt{3}$
- $2\sqrt{5}$ and $-4\sqrt{5}$
- $\sqrt{7}$ and $\sqrt{28}$, because $\sqrt{28} = \sqrt{4 \cdot 7} = \sqrt{4} \cdot \sqrt{7} = 2\sqrt{7}$
- $\sqrt{90}$ and $\sqrt{250}$ because $\sqrt{90} = \sqrt{9 \cdot 10} = \sqrt{9} \cdot \sqrt{10} = 3\sqrt{10}$ and $\sqrt{250} = \sqrt{25 \cdot 10} = \sqrt{25} \cdot \sqrt{10} = 5\sqrt{10}$

To add or to subtract radicals, we need to first simplify radicals, then combine like radicals.

✓ Example 11.3

Add or Subtract radicals:

- $\sqrt{160} + \sqrt{490} = \sqrt{16 \cdot 10} + \sqrt{49 \cdot 10} = \sqrt{16} \cdot \sqrt{10} + \sqrt{49} \cdot \sqrt{10} = 4 \cdot \sqrt{10} + 7 \cdot \sqrt{10} = (4 + 7)\sqrt{10} = 11\sqrt{10}$
- $2\sqrt{27} - 5\sqrt{3} = 2\sqrt{9 \cdot 3} - 5\sqrt{3} = 2\sqrt{9} \cdot \sqrt{3} - 5\sqrt{3} = 2 \cdot 3\sqrt{3} - 5\sqrt{3} = 6\sqrt{3} - 5\sqrt{3} = (6 - 5)\sqrt{3} = 1\sqrt{3} = \sqrt{3}$
- $4\sqrt{18} - 7\sqrt{8} - 3\sqrt{1} = 4\sqrt{9 \cdot 2} - 7\sqrt{4 \cdot 2} - 3 \cdot 1 = 4 \cdot 3\sqrt{2} - 7 \cdot 2\sqrt{2} - 3 = 12\sqrt{2} - 14\sqrt{2} - 3 = (12 - 14)\sqrt{2} - 3 = -2\sqrt{2} - 3$

Note: Only "like" roots can be combined.

How do we simplify non-numerical radicals?

Similar to numbers, variables inside the square root that are squared (raised to the power 2) can be simplified. So, $\sqrt{x^2} = x$ in the same way that $\sqrt{8^2} = 8$. So, we need to find as many multiples of variables that are squared:

$$\begin{aligned}
 \sqrt{x^8} &= \sqrt{x^2 \cdot x^2 \cdot x^2 \cdot x^2} \\
 &= \sqrt{x^2} \cdot \sqrt{x^2} \cdot \sqrt{x^2} \cdot \sqrt{x^2} \\
 &= x \cdot x \cdot x \cdot x \\
 &= x^4 \\
 \sqrt{x^5} &= \sqrt{x^2 \cdot x^2 \cdot x} \\
 &= \sqrt{x^2} \cdot \sqrt{x^2} \cdot \sqrt{x} \\
 &= x \cdot x \cdot \sqrt{x} \\
 &= x^2\sqrt{x}
 \end{aligned}$$

If you have a number and a variable inside the square root (or more than one variable), you work with each one separately. For example:

$$\sqrt{50x^8} = \sqrt{50} \cdot \sqrt{x^8} = 5\sqrt{2} \cdot x^4 = 5x^4\sqrt{2}$$

✓ Example 11.4

Simplify each of the following radical expressions:

$$\begin{aligned}
 \sqrt{y^4 x^8} &= \sqrt{y^4} \cdot \sqrt{x^8} = \sqrt{y^2 \cdot y^2} \cdot \sqrt{x^8} \\
 \text{a) } &= \sqrt{y^2} \cdot \sqrt{y^2} \cdot \sqrt{x^8} = y \cdot y \cdot \sqrt{x^8} = y^2 \cdot \sqrt{x^8} \\
 &= y^2 \cdot x^4 \\
 &= y^2 x^4 \\
 \sqrt{200 \cdot m^4} &= \sqrt{200} \cdot \sqrt{m^4} = \sqrt{100 \cdot 2} \cdot \sqrt{m^4} \\
 \text{b) } &= \sqrt{10^2 \cdot 2} \cdot \sqrt{m^4} = \sqrt{10^2} \cdot \sqrt{2} \cdot \sqrt{m^4} = 10 \cdot \sqrt{2} \cdot m^2 \\
 &= 10 \cdot m^2 \cdot \sqrt{2} = 10m^2\sqrt{2} \\
 m^3 \cdot \sqrt{200 \cdot m^4} &= m^3 \cdot \sqrt{200} \cdot \sqrt{m^4} = m^3 \cdot \sqrt{100 \cdot 2} \cdot \sqrt{m^4} \\
 \text{c) } &= m^3 \cdot \sqrt{10^2 \cdot 2} \cdot \sqrt{m^4} = m^3 \cdot \sqrt{10^2} \cdot \sqrt{2} \cdot \sqrt{m^4} \\
 &= m^3 \cdot 10 \cdot \sqrt{2} \cdot m^2 = 10m^5 \cdot \sqrt{2} \\
 &= 10m^5\sqrt{2} \\
 2\sqrt{63x^3} &= 2 \cdot \sqrt{63} \cdot \sqrt{x^3} = 2 \cdot \sqrt{9 \cdot 7} \cdot \sqrt{x^3} \\
 \text{d) } &= 2 \cdot \sqrt{3^2 \cdot 7} \cdot \sqrt{x^2 \cdot x} = 2 \cdot \sqrt{3^2} \cdot \sqrt{7} \cdot \sqrt{x^2} \cdot \sqrt{x} \\
 &= 2 \cdot 3\sqrt{7} \cdot x \cdot \sqrt{x} = 6x \cdot \sqrt{7x} = 6x\sqrt{7x}
 \end{aligned}$$

Similar to the product rule, the quotient rule allows us to separate square roots as follows:

✚ Quotient Rule for Radicals

$$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$$

And notice that both rules can be read from right to left as follows:

$$\begin{aligned}
 \sqrt{a} \cdot \sqrt{b} &= \sqrt{a \cdot b} \\
 \frac{\sqrt{a}}{\sqrt{b}} &= \sqrt{\frac{a}{b}}
 \end{aligned}$$

This provides us with the necessary tools to combine and simplify square roots, when the operation is **multiplication** or **division** only.

✓ Example 11.5

Simplify completely:

$$\begin{aligned}
 \frac{\sqrt{15}\sqrt{70}}{\sqrt{5}} &= \frac{\sqrt{15 \cdot 70}}{\sqrt{5}} \\
 \text{a. } &= \sqrt{\frac{15 \cdot 70}{5}} = \sqrt{\frac{5 \cdot 3 \cdot 7 \cdot 5 \cdot 2}{5}} \\
 &= \sqrt{5 \cdot 3 \cdot 2 \cdot 7} = \sqrt{210} \\
 -x\sqrt{12y^3} \cdot 3y^2\sqrt{15x} &= -3xy^2 \cdot \sqrt{12y^3 \cdot 15x} = -3xy^2 \cdot \sqrt{12 \cdot 15xy^3} \\
 &= -3xy^2 \cdot \sqrt{4 \cdot 3 \cdot 3 \cdot 5xy^3} \\
 \text{b. } &= -3xy^2 \cdot \sqrt{4 \cdot 3^2 \cdot 5xy \cdot y^2} \\
 &= -3xy^2 \cdot \sqrt{4} \cdot \sqrt{3^2} \cdot \sqrt{5xy} \cdot \sqrt{y^2} \\
 &= -3xy^2 \cdot 2 \cdot 3 \cdot \sqrt{5xy} \cdot y \\
 &= -18xy^3 \cdot \sqrt{5xy}
 \end{aligned}$$

[? Exit Problem](#)

Simplify: $5\sqrt{24} - 2\sqrt{54} - 3\sqrt{16}$

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